

INFO 5653

Text Mining

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Data Scientist (48.0 / 100)	39.8 / 100

40

CURRENT AVG SCORE
across 6 job roles

83

POTENTIAL SCORE
with recommended additions

+52

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces students to techniques for extracting information from text data, including text gathering and cleaning, and text processing methods, such as dimensionality reduction, normalization, and text visualization. It will introduce applications and methodologies of machine learning for text, such as sentiment analysis, association rule mining, and topic modeling.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	48.0	100.0	+52.0	HIGH
MLOps Engineer	45.4	88.7	+43.3	HIGH
Machine Learning Engineer	44.8	77.1	+32.3	HIGH
AI Engineer	41.6	87.7	+46.1	HIGH
Applied Scientist	30.5	75.9	+45.4	HIGH
ML Engineer	28.5	68.7	+40.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 5 of 6 roles

systems

Business Communication 1 of 6 roles

insights

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 5 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: insights.

3 **LOW** **Optional: Feature Engineering** — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

Data Scientist

48.0 → 100.0 (+52.0 pts)

sets product mitigation insights different social build problems

MLOps Engineer

45.4 → 88.7 (+43.3 pts)

detection systems preprocessing tools models

Machine Learning Engineer

44.8 → 77.1 (+32.3 pts)

web sets tools science models technical systems build

AI Engineer

41.6 → 87.7 (+46.1 pts)

models build technical systems problems detection product new

Applied Scientist

30.5 → 75.9 (+45.4 pts)

literature problems models technical systems regulatory fundamental science

ML Engineer

28.5 → 68.7 (+40.2 pts)

systems models build problems product new technical building

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.